Backend & System Engineer

ALEXIS REYNOUARD — Available in September 2027

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🇫🇷 French (native) • 🇺🇸 English (C1 listening, reading, and writing, B2 speaking)

Final-year PhD candidate in formal methods and controller synthesis, seeking **backend and systems engineering roles:** designing, building, and operating reliable systems.

Successfully migrated a hospital's access-control system (5,000+ access badges), increased university lab attendance from 10% to 70%, and built & maintained infrastructure for a non-profit over 10 years.

Specializes in C and C++, Linux systems, automation, and reliability, with a keen interest in integrating safety with AI tools into real-world workflows and exploring embedded systems and OT.

Experience

Odoo Developer & Monitoring Integrations — *Open DB Team* 2015 – 2018

Developed internal **Odoo extensions**, installed, configured and integrated **server monitoring** solutions, completed PostgreSQL administration training.

› `Nagios • Odoo • ERP • Python • DBA • SMS notifications`

Embedded Developer — *Corona Print* 2019, 2024, 2026

Programmed **smart card readers** enabling safe hospital access-control migration (5,000 employees).
Rehired twice for similar successful one-time missions.

› `Smart cards • Embedded C`

IT Technician — *Non-Profit (Volunteer)* 2016 – current

Architected and maintained **custom full-stack CRM** and Linux server (mail server, DB, containerized services), over 10 years.
Managed requirements analysis and deployment of applications, improving operational efficiency.

› `System Administration • Requirement Analysis •`
`Software Architecture • OpenDocument • Multimodal Direct Mailing`

Teaching Assistant — *Université libre de Bruxelles* 2021 – current

Boosted Systems Administration lab attendance from 10% to 70% by redesigning **practical labs**.
Led student teams through conflicts, provided custom **management tools**, performed **code review**, enabling successful project delivery and good grades.
Created, improved, and delivered labs covering Operating Systems (bash, C), System Administration, C++, Software Architecture, Git, and Project Management. Occasionally led labs on Java, Agile Methods, and Network.
Always get very good reviews from anonymous evaluation by students.

› `Conflict Management • Teaching • Project Management`

Junior Researcher — *Université libre de Bruxelles* 2019 – current

Developed theoretical foundations and verification tools for controller synthesis, enabling formal proofs of correctness and safety for real-time systems.

› `Research • Presentation • Problem-Solving • Algorithm Design •`
`Low-Level Programming • Formal Methods`

Education — *Université libre de Bruxelles*

Ph.D. Computer Science – Formal Methods & Controller Synthesis 2019 – 2027 (expected)
Master in Computer Science ("Grande Distinction"/With Honors) 2017 – 2019

Publications

The Non-Cooperative Rational Synthesis Problem for Subgame Perfect Equilibria and omega-regular Objectives, *Véronique Bruyère, Jean-François Raskin, A. R., Marie van den Bogaard,* CONCUR 2025.

A Zone-Based Algorithm for Timed Parity Games, *Gilles Geeraerts, Frédéric Herbreteau, Jean-François Raskin, A. R.,* FSTTCS 2025.

Interests

Automation • Classification • Technologies • Physics
Popularization • Education • Teaching • Pedagogy
Typography • Design • Photography
History • Cultures • Philosophy
Languages • Etymology
Walking • Cycling

Skills

C++ • C • Python • Linux System Administration
Backend Development • Containerization (Docker, Podman) • PostgreSQL / SQLite / SQL
Performance profiling & Optimization • Automation Scripting (Bash, Python)

Research • Formal Verification / Formal Methods

Communication • Teaching • Conflict Management • Project Management

European First Aid Certificate